

Daniel Banariba

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Systems Engineer and Data Engineer. I build production data pipelines with Dagster, Python and dbt over a SQL Server DWH, and MCP tooling that lets AI operate real infrastructure — all under strict TDD. Cut a monthly forecast from ~18h to ~61 min with byte-identical output.

WORK EXPERIENCE

Grupo Farinter

RPA Developer & Data Engineer | November 2025 - Present

- Optimized a monthly demand forecast (~23,000 series) from ~18 hours to ~61 minutes in production, parallelizing with ProcessPoolExecutor and pruning dead series, with output byte-identical to the sequential baseline and deterministic. Delivered under strict TDD (30 tests) with no new dependencies.
- Turned operating production infrastructure —orchestration, database queries and deployments— into something handled from an AI assistant, building a fleet of 6 self-built MCP (Model Context Protocol) servers with a read-only validator, two-step confirmation and ~10-20x payload reduction on wide reads.
- Cut false positives in the credit portfolio's compliance control by 94%, building the RPA that screens it against international restricted lists with fuzzy matching, without losing a single true positive. Validated and signed off by the Compliance team.
- Automated the daily reconciliation between the payments system and the bank, integrating its H2H REST API (OAuth2 with credentials in HashiCorp Vault) with an incremental dbt model and 4 data-quality asset checks; 9 chained PRs with 1,524 tests.
- Recovered a CRM ingestion that was silently dropping most records while every run stayed green; reloaded millions of rows with no duplicates and, by measuring dlt worker contention, raised load throughput from 91 to 368 rows/s (4x).
- Brought security auditing into the delivery cycle for production applications: column-level access control over sensitive figures, credential-exposure review in logs, and input validation on file ingestion, with every finding remediated and covered by regression tests.
- Migrated the Power BI failure monitor from a standalone script to a production asset against its REST API with OAuth2 over Microsoft Entra ID, cutting the run from ~11 to ~3.4 minutes (3.2x) by reusing the HTTP session.
Technologies: Python, Dagster, SQL Server, PostgreSQL, dbt, Polars, Docker, HashiCorp Vault, MCP, GraphQL, OAuth2, pytest, TDD, SDD

Analiza Clinical Laboratories

Software Developer | March 2025 - October 2025

- Automated the VITEK-CRM integration with FastAPI and ASTM/HL7 protocols in production for bidirectional processing of culture results, with message traceability, automatic retries and logging.
- Modeled clinical catalogs with SQLAlchemy ORM (7,429 microorganisms, 278 antibiotics) with complex relationships between antibiogram headers and details.
- Designed and deployed corporate proxy infrastructure with Docker, Squid and OpenVPN connecting operations across Honduras, Guatemala and El Salvador, reducing management of 30+ machines to a single point.
- Implemented a referral commission system that automates the calculation on referred laboratory services, with validations preventing duplicates and safeguarding financial transaction integrity.
Technologies: Python, FastAPI, TypeScript, NestJS, PostgreSQL, SQLAlchemy, Docker, Websocket, ASTM, HL7, VITEK, OpenVPN

GuabaBIT

QA & Backend Developer | October 2024 - March 2025

- Designed the technical architecture documentation for a community payments system with RESTful services, NoSQL databases (DynamoDB) and AWS services (S3, SNS).
- Built the real-time notifications feature —endpoints, models and backend services— letting users receive instant alerts on relevant updates.
- Implemented E2E tests with Playwright for critical flows (login, forms, quotes, handshake) and automated API testing with Postman/Apidog, building reusable collections and test scripts, and documenting the APIs with Swagger for maintainability.

- Wrote and executed manual test cases in Jira to verify system integrity, reporting and documenting critical bugs with precise reproductions.

Technologies: TypeScript, NestJS, DynamoDB, AWS (S3, SNS), Playwright, Postman, Apidog, Swagger, Jira, Microservices

SKILLS

- Languages: Python, TypeScript, JavaScript, C++, Rust, SQL, T-SQL
- AI & AI-Assisted Development: Claude Code, Codex CLI, OpenCode, MCP (Model Context Protocol), Agent Skills, AI Agents, Sub-agents, Agent Orchestration, Prompt Engineering, SDD, Anthropic API, LLMs / GenAI
- Frameworks & Orchestration: Dagster, FastAPI, NestJS, Reflex, Astro
- Databases: SQL Server, PostgreSQL, MongoDB, DynamoDB, SQLite
- Cloud & Infrastructure: Microsoft Azure, AWS (S3, SNS, DynamoDB), Terraform, Docker, CI/CD, GitHub Actions, HashiCorp Vault, nginx, Dokploy
- Testing & QA: pytest, strict TDD, Playwright, Mutation Testing, Jest, Postman, Jira
- Data Engineering: ETL/ELT, Data Warehousing, Data Modeling, Data Quality, dbt, Data Orchestration (Dagster), Polars, dlt, Power BI, statsforecast

PROJECTS

Accounting System — Azure Infrastructure with Terraform

Infrastructure as code for a secure, scalable accounting system on Azure: virtual network, SQL database, storage and web app, with advanced security configuration. Public repository at github.com/danielbanariba/sistema-contable.

Technologies: Azure, Terraform, Azure Virtual Network, Azure SQL Database, Azure App Service, IaC

Competitive Pricing Intelligence — Analytics Dashboard

Full-stack Python (Reflex) dashboard with 10 analytical views: margin erosion detection via SCD2 history, suggested-price simulation and column-level access control over sensitive figures. Refactored to multi-country, with a second country in production. 1,790 tests and CI/CD.

Technologies: Python, Reflex, SQL Server, dbt, SCD2, Playwright, Docker, nginx, CI/CD

hvault — TLS Certificate Sync in Rust

Pure Rust service (CLI + daemon) that syncs TLS certificates across nodes using HashiCorp Vault (KV-v2) as the source of truth, with atomic writes for hot-reload without downtime, x509 validation and a Leptos/WASM admin UI with OIDC.

Technologies: Rust, HashiCorp Vault, Traefik, Leptos, WASM, OIDC, x509, Docker, GitHub Actions

vk3d-proton — Open Source Contribution

Merged PR in the D3D12→Vulkan translation layer Proton uses to run Windows games on Linux: the ray tracing pipeline failed when no explicit root signature was bound, and I fixed it by binding it implicitly from the pipeline state object itself.

Technologies: C, Vulkan, Direct3D 12, Graphics Pipeline, Open Source

EDUCATION

National Autonomous University of Honduras

B.S. in Mathematics (in progress) | Tegucigalpa, Honduras | 2024 - Present

National Autonomous University of Honduras

B.S. in Computer Systems Engineering | Tegucigalpa, Honduras | 2019 - 2025

CERTIFICATIONS

- Oracle Next Education F2 T5 Back-end - Alura Latam, 2023
- Database Foundations - Oracle Academy, 2023

LANGUAGES

- Spanish: Native
- English: Technical reading and comprehension; interviews and work in Spanish